

Cox Proportion Hazards Regression

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Multivariable

Include multiple independent variables

Estimate Effect Size

Hazard Ratio = Treatment Hazard / Control Hazard

Hazard: Probability of an event at any given point in time

Demo

Cox Proportion Hazards Regression

Knowledge Check

True or False: According to the model, being diagnosed with depression is more hazardous than not being diagnosed with depression.

Call:

```
coxph(formula = Surv(mos_to_death, died) ~ female + age_2014 +  
      bp_dx + depress_dx, data = hrs_df)
```

	coef	exp(coef)	se(coef)	z	p
female	0.393772	1.482562	0.086492	4.553	5.3e-06
age_2014	0.290459	1.337042	0.009521	30.508	< 2e-16
bp_dx	-0.058589	0.943094	0.089292	-0.656	0.511726
depress_dx	0.375581	1.455837	0.103778	3.619	0.000296

Likelihood ratio test=1063 on 4 df, p=< 2.2e-16
n= 2355, number of events= 660
(62 observations deleted due to missingness)

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<https://pollev.com/albertlee033>

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Proportional Hazards Assumption

HRs are constant over time

Test

Null Hypothesis: HRs are constant over time

Alternative Hypothesis: HRs are not constant over time

Demo

Test the Proportional Hazards
Assumption of a Cox Regression
Are HRs constant over time?

Knowledge Check

True or False: Our model meets the proportional hazards assumption overall

	chisq	df	p
female	0.00396	1	0.950
age_2014	4.33717	1	0.037
bp_dx	0.45907	1	0.498
depress_dx	0.00075	1	0.978
GLOBAL	5.74540	4	0.219

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